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Abstract

This analytical study investigates how a Procurement Recommendation Score (PRS) can improve supplier evaluation in civilian supply chains. The PRS is defined as the product of Supplier Performance Score (SPS) and Product Criticality Score (PCS). This metric has been calculated by taking inspiration from the U.S. Department of Defense's Supplier Performance Risk System (SPRS), where supplier performance as well as product criticality is taken into consideration to choose the best supplier for a particular contract. The PRS framework takes into consideration multiple procurement risk dimensions which are compliance, defect rate, cost efficiency, frequency of negligence, and rate of cancellation. These variables will help generate a consolidated supplier score. Using a publicly available procurement dataset, SPS and PCS are computed and then combined to calculate PRS. This score then enables structured, risk-adjusted allocation of suppliers to product categories.

To extend the analytical contribution, predictive models are devised to estimate likelihood of compliance and predict defect rates at order level for each suppliers. Logistic regression predicts compliance flags, while a Random Forest regression model is used in forecasting defect rates. The predictive insights from these analytical runs support prescriptive recommendations such as safety-stock adjustments and supplier-specific mitigation strategies.

The results validate that PRS provides a transparent, data-driven mechanism suitable for supplier selection and enhances procurement decision-making by integrating past performance, product risk, and predicted operational outcomes. The study highlights practical implications for reducing supply disruptions and improving sourcing resilience in civilian organizations by taking inspiration from military supply chains.

Nevertheless, the analysis is limited by a small supplier dataset, subjective performance weightings, and modelling challenges that arised from class imbalance of the dataset considered.

Introduction

Modern supply chains operate in an increasingly complex, high-stakes landscape that is further complicated by supplier unreliability, quality failures, and long replenishment cycles. All those factors which can directly affect operational continuity (Hallikas & Lintukangas 2016). Traditional methods of supplier evaluation rely mostly on descriptive historical KPIs, which to an extent gives idea about the supplier's capability based on their historical performances but offer limited information to take forward-looking decisions (Zsidisin et al. 2004).

As a result, civilian supply chains continue to face challenges in allocating the right suppliers to the right product categories, especially when product importance, risk exposure, and operational criticality vary significantly across different product categories.

To address these limitations and to provide better predictive insights, this study introduces a Procurement Recommendation Score (PRS). The PRS framework is designed to strengthen supplier selection decisions. The model combines two key variables which are Supplier Performance Score (SPS) and Product Criticality Score (PCS). The SPS in turn is derived from key procurement KPIs such as defect rate, compliance flag, cost efficiency, pattern of negligence, and cancellation rate. The PCS captures the importance of each category in optimal operational performance. This structure is developed drawing inspiration from the U.S. DoD's Supplier Performance Risk System (SPRS) (Department of the Navy, 2024). The PRS framework adapts the risk-weighted principles of SPRS to the context of the civilian supply chain.

In addition to descriptive performance evaluation, this research also incorporates predictive analytics to help anticipate the likelihood of compliance and defect rates in future orders. These predictive components provide additional insights into supplier risk exposure and enable prescriptive recommendations to prevent disruptions. Collectively, the SPS, PCS, PRS, and predictive models form an integrated analytical framework that helps make better sourcing decisions and enhances supply assurance by reducing operational risks in civilian supply chains.

Research Question

How can an SPRS-inspired Procurement Recommendation Score (PRS), constructed from supplier performance metrics and product criticality and strengthened through predictive modelling of compliance and defect rates, be used to improve supplier selection in civilian supply chains?

Systematic Literature Review

Purpose of the Review

The objective of this review is to evaluate how existing supplier evaluation and risk-assessment techniques support strategic supplier selection in civilian supply chains. By systematically analyzing current performance models, this literature review identifies conceptual and methodological gaps that motivate the design of the Procurement Recommendation Score (PRS).

Search Strategy

A structured search was conducted across multiple sources that include Google Scholar, Scopus, and IEEE Xplore between 2000–2024. The following keyword combinations were used:

- *“supplier evaluation model”, “supplier performance measurement”*
- *“supply risk assessment”, “supplier scoring framework”*
- *“procurement KPIs”, “supplier selection criteria”*
- *“product criticality”, “item risk classification”*
- *“Kraljic matrix AND supplier selection”*
- *“SRM AND risk”*

Inclusion and Exclusion Criteria

Inclusion:

- Peer-reviewed journals or reputable conference publications
- Studies proposing supplier evaluation models, risk assessment tools, or procurement KPIs
- Literature discussing item criticality or supplier–product allocation
- Publications offering conceptual or empirical evaluation frameworks

Exclusion:

- Papers focused solely on production planning or inventory optimization without supplier assessment
- Non-English publications

Thematic Synthesis of Findings

Supplier Evaluation Criteria

Multiple studies emphasize that the traditional supplier evaluation frameworks are heavily weighted toward quantitative KPIs such as cost, quality, and delivery (Zsidoisin et al. 2004; Er Kara & Fırat 2018, Namdar et al. 2018, Pereira, Christopher & Da Silva 2014, Blackhurst, Scheibe & Johnson 2008). However, these models do not sufficiently consider behavioral or risk-related dimensions, nor do they consider how supplier performance interacts with product criticality.

Risk Assessment Frameworks

The literature emphasizes established frameworks such as the Kraljic Matrix which classify products by risk and importance (Garzon et al. 2019), and SPRS, which evaluates suppliers using price, item, and vendor risk dimensions (Department of the Navy, 2024). Even though SPRS combines these two dimensions in the case of military supply chains, similar implementation is rare in the case of civilian counterparts. Further, the lack of linkage between evaluation criteria and a firm's supply chain strategy also makes the results of existing frameworks highly unreliable (Xia, Ma & Lim 2008).

Supplier Relationship Management (SRM) and Resilience

Literature on SRM underlines the importance of continuous monitoring, segmentation, and predictive assessment of suppliers to strengthen resilience (Hallikas & Lintukangas 2016; Pereira et al. 2014, Vaka 2024). However, existing models do not discuss how supplier performance should change/influence sourcing decisions across different product categories.

Identified Gaps and Motivation for PRS

Across the reviewed studies, several limitations consistently emerge:

Fragmentation between supplier evaluation and item criticality

Existing models evaluate suppliers for products but rarely integrate them in the case of civilian supply chains.

Lack of risk-adjusted supplier–product allocation mechanisms

None of the frameworks in the reviewed literature multiplies supplier behaviour KPIs with product criticality levels to guide strategic sourcing.

Insufficient predictive capability

Descriptive KPIs dominate the literatures, and only some models use predictive analytics in combination with these KPIs (e.g., compliance or defect risk forecasting).

Operationalization gap for civilian supply chains

SPRS is effective in defense procurement, but there is limited translation to civilian environments, and the implications need to be studied.

These gaps justify the development of a Procurement Recommendation Score (PRS) which serves as a unified metric that combines supplier performance (SPS) with product criticality (PCS) and incorporates the insights obtained from predictive modelling to help make better sourcing decisions.

Methodology

This study follows a multi-stage analytical process that comprises data cleaning, development of appropriate KPIs, supplier scoring, product criticality modelling, predictive analytics, and finally, prescriptive recommendations. The methodology draws inspiration from SPRS and is designed to move from descriptive to predictive and prescriptive decision support for civilian supply chains

The PRS ranks supplier–product pairs, and predictive modelling further enhances this by identifying the expected future risk of each supplier beyond historical averages.

Data Collection and Preparation

The dataset used for this study was a public procurement KPI dataset available on Kaggle. The dataset contains 777 rows and 10 variables of transaction-level information on purchase orders, negotiated prices, quantities, delivery dates, item categories, defective units, and compliance indicators identified by unique purchase order ID. Additional features such as lead time, compliance flag, and category-specific criticality values were defined externally to construct analytical variables which was necessary for modelling.

The following data preprocessing steps were conducted:

- Separating orders labelled as cancelled was moved to a “cancellation dataset” to calculate cancellation rate
- Separating orders with missing delivery dates into a “negligence dataset” for supplier risk assessment
- Computing lead time for valid completed orders using the formula $\text{Delivery_Date} - \text{Order_Date}$
- Ensuring numeric fields (Quantity, Negotiated_Price, Lead_Time, Defective_Units) contained no missing values before initiating modelling

This preprocessing helped create a consolidated modelling DataFrame (model_df) containing all necessary features for descriptive and predictive analytics.

KPI Development

Five supplier-level KPIs were developed from the data obtained after preprocessing which help to get an idea of the supplier's performance such as quality, reliability, cost performance, documentation adherence, and risk behaviour. These KPIs are crucial to constructing the Supplier Performance Score (SPS).

(a) Supplier Defect Rate

$$\text{Defect Rate} = \frac{\sum \text{Defective Units}}{\sum \text{Quantity}}$$

```
defects = df[df['Order_Status'] == 'Delivered'].groupby('Supplier')['Defective_Units'].sum()
delivered_qty = df[df['Order_Status'] == 'Delivered'].groupby('Supplier')['Quantity'].sum()

supplier_defect_rate = defects / delivered_qty
display(supplier_defect_rate)
```

Supplier	0
Alpha_Inc	0.018153
Beta_Supplies	0.074192
Delta_Logistics	0.106421
Epsilon_Group	0.025771
Gamma_Co	0.044518

dtype: float64

(b) Supplier Cost Efficiency Score

Savings per Unit = Unit Price – Negotiated Price

$$\text{Cost Efficiency} = \frac{\sum (\text{Savings per Unit} \times \text{Quantity})}{\sum (\text{Unit Price} \times \text{Quantity})}$$

```
df['Price_Difference'] = df['Unit_Price'] - df['Negotiated_Price']

cost_efficiency = df.groupby('Supplier')['Price_Difference'].mean()
display(cost_efficiency)
```

Supplier	Price_Difference
Alpha_Inc	4.738369
Beta_Supplies	4.671538
Delta_Logistics	4.337719
Epsilon_Group	4.757229
Gamma_Co	4.642168

dtype: float64

(c) Supplier Compliance Score

Compliance Flag is 1 if all conditions of that particular transaction is met, including delivery due date, 0 if some or none of the conditions are met. Supplier compliance score is the mean compliance across all orders.

```
valid_compliance_df = df[
    (df['Order_Status'] != 'Cancelled') &
    (df['Delivery_Date'].notnull() | (df['Order_Status'] == 'Pending') | (df['Order_Status'] == 'Partially Delivered'))
].copy()
display(valid_compliance_df.head())
```

	PO_ID	Supplier	Order_Date	Delivery_Date	Item_Category	Order_Status	Quantity	Unit_Price	Negotiated_Price	Defective_Units	Compliance	Price_Difference
1	PO-00002	Delta_Logistics	2022-04-25	2022-05-05	Office Supplies	Delivered	1509	39.32	37.34	235.0	Yes	1.98
2	PO-00003	Gamma_Co	2022-01-26	2022-02-15	MRO	Delivered	910	95.51	92.26	41.0	Yes	3.25
3	PO-00004	Beta_Supplies	2022-10-09	2022-10-28	Packaging	Delivered	1344	99.85	95.52	112.0	Yes	4.33
4	PO-00005	Delta_Logistics	2022-09-08	2022-09-20	Raw Materials	Delivered	1180	64.07	60.53	171.0	No	3.54
5	PO-00006	Epsilon_Group	2022-08-17	2022-08-29	MRO	Delivered	1145	69.21	63.57	39.0	Yes	5.64

```
valid_compliance_df['Compliance_Flag'] = valid_compliance_df['Compliance'].map({'Yes': 1, 'No': 0})
display(valid_compliance_df.head())
```

	PO_ID	Supplier	Order_Date	Delivery_Date	Item_Category	Order_Status	Quantity	Unit_Price	Negotiated_Price	Defective_Units	Compliance	Price_Difference	Compliance_Flag
1	PO-00002	Delta_Logistics	2022-04-25	2022-05-05	Office Supplies	Delivered	1509	39.32	37.34	235.0	Yes	1.98	1
2	PO-00003	Gamma_Co	2022-01-26	2022-02-15	MRO	Delivered	910	95.51	92.26	41.0	Yes	3.25	1
3	PO-00004	Beta_Supplies	2022-10-09	2022-10-28	Packaging	Delivered	1344	99.85	95.52	112.0	Yes	4.33	1
4	PO-00005	Delta_Logistics	2022-09-08	2022-09-20	Raw Materials	Delivered	1180	64.07	60.53	171.0	No	3.54	0
5	PO-00006	Epsilon_Group	2022-08-17	2022-08-29	MRO	Delivered	1145	69.21	63.57	39.0	Yes	5.64	1

(d) Supplier Negligence Score

If delivered orders are missing a delivery date, these are considered a fault of the supplier due to some missing paperwork.

$$\text{Negligence Rate} = \frac{\text{Orders with Missing Delivery Date}}{\text{Total Orders}}$$

```
negligence_count = df[df['Order_Status'] == 'Delivered'] \
    .groupby('Supplier')['Delivery_Date'] \
    .apply(lambda x: x.isnull().sum())

total_delivered = df[df['Order_Status'] == 'Delivered'] \
    .groupby('Supplier')['Delivery_Date'] \
    .size()

negligence_rate = (negligence_count / total_delivered).round(3) # 3 decimal places
display(negligence_rate)
display(negligence_count)
```

	Delivery_Date
Supplier	
Alpha_Inc	0.168
Beta_Supplies	0.091
Delta_Logistics	0.142
Epsilon_Group	0.108
Gamma_Co	0.097

dtype: float64

	Delivery_Date
Supplier	
Alpha_Inc	18
Beta_Supplies	10
Delta_Logistics	17
Epsilon_Group	13
Gamma_Co	10

dtype: int64

(e) Supplier Cancellation Rate

$$\text{Cancellation Rate} = \frac{\text{Cancelled Orders}}{\text{Total Orders}}$$

```
cancelled = df[df['Order_Status'] == 'Cancelled'].groupby('Supplier').size()
total_orders = df.groupby('Supplier').size()

supplier_cancellation_rate = cancelled / total_orders
display(supplier_cancellation_rate)
```

Supplier	0
Alpha_Inc	0.092199
Beta_Supplies	0.089744
Delta_Logistics	0.093567
Epsilon_Group	0.084337
Gamma_Co	0.041958

dtype: float64

Normalization and Calculation of Supplier Performance Score (SPS)

To ensure all KPIs operate on a comparable scale, each metric was normalized to the range of [0, 1] using the min-max method:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Negatively oriented KPIs (i.e. lower the rate, better the performance), here, defect rate, negligence, cancellations were inverted:

$$X_{\text{inv}} = 1 - X_{\text{norm}}$$

Then weightage was assigned to each KPIs and the final SPS equation was deduced:

$$\text{SPS} = 0.25(\text{Compliance}) + 0.30(\text{Defect}) + 0.25(\text{Cost Efficiency}) + 0.10(\text{Negligence}) + 0.10(\text{Cancellation})$$

These weightages were decided by risk-based principles inspired by the US DoD's SPRS framework and informed conceptually by Multi-Criteria Decision-Making literature such as the Analytic Hierarchy Process (AHP) discussed in Pamplona (2016). This formula emphasizes quality and reliability as primary drivers of supplier performance risk (Department of the Navy, 2024).

Product Criticality Score (PCS)

There were five product categories in the dataset, and each product category was assigned a criticality value between 1 and 5, based on its operational dependency, lead-time, and impact of failure, taking inspiration from the Kraljic matrix (Garzon et al. 2019).

Category	PCS	Justification
Raw Materials	5	• Considered core input for production • Supply disruption directly halts operations • Has the longest lead time • Is of highest strategic importance
Electronics	4	• High impact on operations • Subject to long lead times, might have to be imported • Failures can stop equipment or processes.
MRO (Maintenance, Repair & Operations)	3	• Important for machine uptime and safety • Moderate operational impact • Can be stocked
Packaging	2	• Crucial for shipping but easily replaced • Low failure consequences • Short lead time.
Office Supplies	1	• Low operational criticality • Easily substitutable • Minimal risk.

Procurement Recommendation Score (PRS)

The PRS model takes into consideration both the supplier capability and product criticality helping stakeholders make better procurement decisions:

$$\text{PRS} = \text{SPS} \times \text{PCS}$$

This risk-adjusted weighting follows the logic of the U.S. DoD's SPRS and that of the Kraljic Purchasing Portfolio Model adapted to match the context of civilian procurement. The multiplicative design ensures that a supplier only receives a high procurement recommendation score when it combines strong historical performance with a product category that is crucial to operations. Hence, high scoring suppliers are considered valuable to the company's sustained operations. PRS can be used to rank supplier-product combinations and to illustrate how sourcing decisions could be re-prioritized under the proposed framework.

Predictive Modelling

Existing models rely mostly on descriptive KPIs to gain inference on supplier performance. The addition of predictive models allows forward-looking supplier risk assessment, which further strengthens numbers obtained through PRS.

By forecasting risks such as non-compliance or potential defects, it allows stakeholders to prepare preventive measures in advance which in turn ensure operational continuity even when disruptions arise. With this in mind, two predictive models were implemented:

(a) Compliance Prediction (Classification Model)

A logistic regression model that predicts the likelihood of compliance of a future order from a supplier. Features included:

- Quantity
- Unit Price
- Negotiated Price
- Price Difference
- Lead Time
- Supplier Identifier
- Item Category

The performance of this model was evaluated using accuracy, precision, recall, and F1 score.

(b) Defect Rate Prediction (Regression Model)

The Random Forest Regression model was used to estimate the defect rate at the order level:

$$\text{Defect Rate} = \frac{\text{Predicted Defective Units}}{\text{Quantity}}$$

Prescriptive Layer

The predictive insight enables stakeholders to make prescriptive decisions regarding buffer-stock, further enabling procurement professionals to order the required extra units when defect risk is high, and cost conditions allow.

The predicted defect rate that is suitable for operational recommendations is calculated as:

$$\text{Recommended Buffer} = \text{Predicted Defect Rate} \times \text{Order Quantity} \times \text{Safety Factor}$$

This method supports proactive sourcing strategies that will help prevent stockouts and production disruptions. The safety factor is unique to an organization and depends on the lead time and availability of that product.

Data analysis

Data analysis of dataset was conducted in two stages: descriptive scoring analysis and predictive modelling.

Descriptive Scoring Analysis

Data Cleaning and Pre-Processing

The initial step involved scanning the dataset for missing values using `df.isnull().sum()`. Null entries were found primarily in `Delivery_Date` and `Defective_Units`. These were not removed but interpreted according to business logic:

Delivery_Date = Null

- For cancelled orders, this is expected and therefore retained.
- For delivered orders, a missing delivery date indicates a process negligence on the supplier side.

Defective_Units Null

- For delivered orders with null defects, the defect count was assumed to be zero.
- For cancelled orders, this field was ignored as defects are irrelevant.

This treatment ensured no artificial bias was introduced through unnecessary row deletion while still preserving operational meaning.

```
import pandas as pd

df = pd.read_csv(path)
display(df.isnull().sum())
```

	0
PO_ID	0
Supplier	0
Order_Date	0
Delivery_Date	87
Item_Category	0
Order_Status	0
Quantity	0
Unit_Price	0
Negotiated_Price	0
Defective_Units	136
Compliance	0

dtype: int64

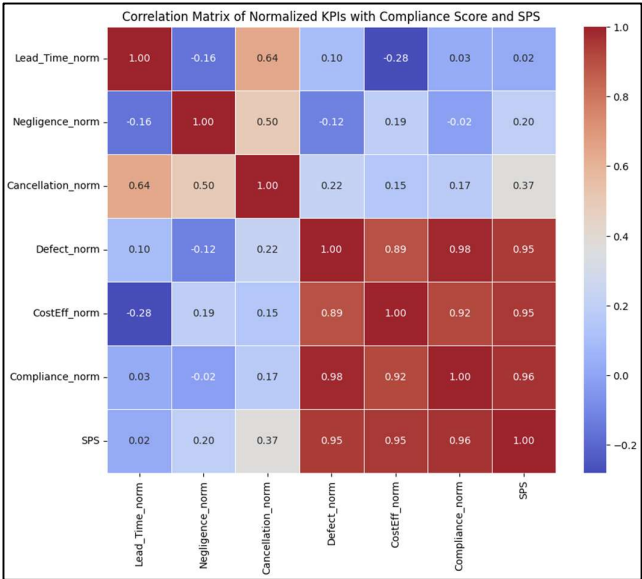
The **lead time** was then computed as the number of days between Order_Date and Delivery_Date. Because the dataset does not provide a contractual due date, a formal “Delivery Score” cannot be computed. Instead, deliveries marked “Delivered” and have compliance yes, were treated as orders meeting the due date, forming the foundation for later compliance scoring.

Following the initial data preparation, a series of descriptive analyses were performed to understand the underlying characteristics of the dataset before moving to KPI engineering. The following numerical variables were explored: Lead Time, Quantity, Negotiated Savings (Price Difference), and Defective Units. Summary tables were generated to obtain key measures such as mean, median, standard deviation, minimum and maximum values, and the interquartile range (IQR). These statistics helped establish the central tendency and dispersion of each operational metric.

```
display(model_df.describe().T)
```

	count	mean	std	min	25%	50%	75%	max
Quantity	492.0	1101.036585	620.432103	51.00	655.5000	1103.500	1545.0000	5000.00
Unit_Price	492.0	58.271057	28.139169	11.28	33.3800	57.970	83.0275	108.75
Negotiated_Price	492.0	53.782398	26.294277	9.98	30.6125	52.365	77.1900	106.30
Defective_Units	402.0	74.427861	68.642236	0.00	26.0000	49.000	100.0000	308.00
Price_Difference	492.0	4.488659	3.324389	0.18	1.9325	3.585	6.1650	15.56
PCS	492.0	2.951220	1.393050	1.00	2.0000	3.000	4.0000	5.00
Lead_Time	492.0	10.491870	5.781311	-5.00	5.0000	11.000	15.0000	20.00
Compliance_Flag	492.0	0.825203	0.380180	0.00	1.0000	1.000	1.0000	1.00

A deeper inspection of lead time revealed there is substantial variation among the performance of suppliers. The IQR and maximum values highlighted a subset of orders with abnormally high lead times, which might be pointing to potential bottlenecks. Similarly, the distribution of defective units was right-skewed, with most orders showing low or zero defects, but a small number of these had unusually high defect counts. This skewness is typical in quality datasets and supports the need for KPI normalization.

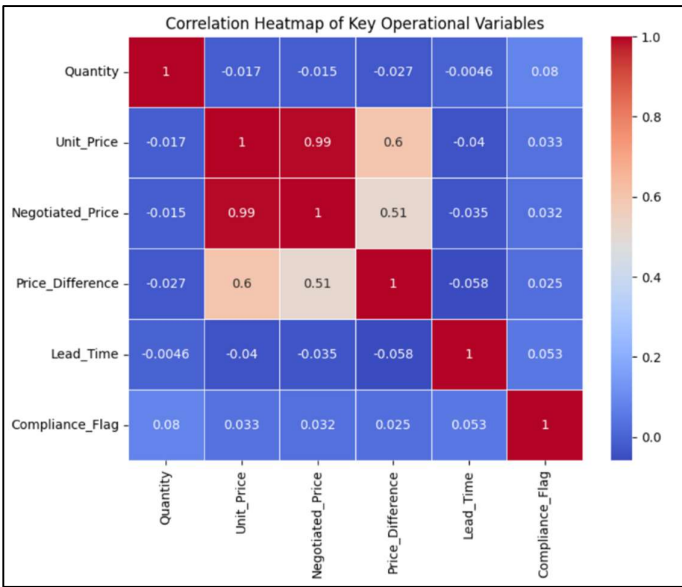


Following the data processing, the KPIs were devised as mentioned in the methodology section for developing the PRS framework.

Following KPI engineering, to examine structural relationships among the operational variables, a correlation heatmap was generated in python. This visual helps identify whether order-level factors such as quantity, price behavior meaningfully explain compliance outcomes or defect behavior.

The heatmap suggests that in this case the compliance flag is largely uncorrelated with operational variables which in turn highlight the need to incorporate SPS KPIs and predictive modelling to better understand compliance behavior of suppliers.

Another correlation matrix that involves all the KPIs and the SPS scores was created. This shows that SPS is very strongly correlated with Defect_norm (0.95), CostEff_norm (0.95), and Compliance_norm (0.96). This pattern is expected because SPS is mathematically derived from these underlying KPIs, and therefore strong positive associations indicate that the scoring framework is functioning as designed. The high correlations confirm the internal consistency of the KPI structure, i.e. suppliers with better defect performance, and higher compliance naturally achieve higher SPS values. This validation step reinforces that the SPS calculation reliably captures the intended dimensions of supplier performance.



Predictive Modelling

The predictive modelling component aimed to determine whether two key procurement outcomes, compliance and defect, rate can be predicted from supplier/operational variables. For this, two supervised machine learning models were used. Logistic Regression was used for classification and Random Forest Regression was deployed for continuous prediction. Both models were trained after removing rows with missing compliance or lead time values to ensure clean inputs for modelling.

Classification Model – Predicting Order Compliance (Logistic Regression)

The goal was to predict whether a future order will be compliant based on operational features such as supplier, product category, supplier pricing behavior, and lead time and ultimately, the hypothesis was devised:

H₀: Supplier-identity has no significant influence on compliance outcomes.

H₁: Supplier-identity significantly influence compliance outcomes.

Supplier identity serves as a proxy for underlying supplier performance patterns captured through KPIs in both predictive models.

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# 1. Classification Target
y1 = model_df['Compliance_Flag']

# 2. Features
X1 = model_df[['Quantity', 'Unit_Price', 'Negotiated_Price', 'Price_Difference',
               'Lead_Time', 'Supplier', 'Item_Category']]

# 3. Preprocessing (encode categorical)
categorical_cols = ['Supplier', 'Item_Category']
numeric_cols = ['Quantity', 'Unit_Price', 'Negotiated_Price', 'Price_Difference', 'Lead_Time']

preprocessor = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_cols),
        ('num', 'passthrough', numeric_cols)
    ]
)

# 4. Build model pipeline
clf_model = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(max_iter=1000)) # Increased max_iter to address convergence warning
])

# 5. Train/test split
X1_train, X1_test, y1_train, y1_test = train_test_split(X1, y1, test_size=0.2, random_state=42)

# 6. Train
clf_model.fit(X1_train, y1_train)

# 7. Predict
y1_pred = clf_model.predict(X1_test)

# 8. Evaluate
print("Accuracy:", accuracy_score(y1_test, y1_pred))
print("Confusion Matrix:\n", confusion_matrix(y1_test, y1_pred))
print("Classification Report:\n", classification_report(y1_test, y1_pred))
```

Logistic Regression was selected as the classification model because the target variable, order compliance, is binary. The model also offers easily interpretable coefficients which enables clear insights on how operational factors such as lead time, pricing behaviour, and supplier characteristics influence the likelihood of compliance in future orders.

As a part of data preparation, categorical variables such as Supplier and Item Category were transformed using one-hot encoding to ensure they could be effectively interpreted by the classification model. As for the numeric variables (Quantity, Unit Price, Price Difference, and Lead Time), they were retained as such. To maintain analytical integrity, rows that had missing value in either the Compliance_Flag or Lead_Time fields were removed for this modeling, as these missing values would compromise the reliability and accuracy of the model training process.

After cleaning and preparing the dataset, an 80/20 train–test split was applied to allow the model to learn from the majority of the data while reserving a separate subset for unbiased evaluation of predictive performance.

Model evaluation was conducted using accuracy, precision, recall, and F1-score.

Regression Model – Predicting Future Defect Rates (Random Forest)

The goal was to predict the Defect Rate of future orders using historical data that involves supplier behaviour, product category, pricing variables, and lead time and ultimately, the following hypothesis was devised:

H₀: Operational factors contribute more to predicting defect rate than supplier identity.

H₁: Supplier identity contributes more predictive power to defect-rate outcomes compared to operational variables.

A Random Forest Regressor was selected for predicting defect rates because it effectively captures non-linear interactions among operational variables, which are common in quality-related outcomes. The algorithm is also capable of handling mixed feature types (categorical as well as numerical) making it well-suited to procurement data that includes supplier identities, product categories, and quantitative measures. Additionally, Random Forest models generally provide greater accuracy and stability when compared to linear regression, especially when dealing with irregular or noisy patterns. Thus, making them an appropriate choice for forecasting defect behavior in this context.

```

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# 1. Regression Target
y2 = model_df_reg['Defect_Rate']

# 2. Features
X2 = model_df_reg[['Quantity', 'Unit_Price', 'Negotiated_Price', 'Price_Difference',
                  'Lead_Time', 'Supplier', 'Item_Category']]

# 3. Preprocessing (same as before)
preprocessor2 = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore'), ['Supplier', 'Item_Category']),
        ('num', 'passthrough', ['Quantity', 'Unit_Price', 'Negotiated_Price', 'Price_Difference', 'Lead_Time'])
    ]
)

# 4. Regression pipeline
reg_model = Pipeline(steps=[
    ('preprocessor', preprocessor2),
    ('regressor', RandomForestRegressor(n_estimators=200, random_state=42))
])

# 5. Train/test split
X2_train, X2_test, y2_train, y2_test = train_test_split(X2, y2, test_size=0.2, random_state=42)

# 6. Train
reg_model.fit(X2_train, y2_train)

# 7. Predict
y2_pred = reg_model.predict(X2_test)

# 8. Evaluation
print("MAE:", mean_absolute_error(y2_test, y2_pred))
print("R2 Score:", r2_score(y2_test, y2_pred))

```

As a part of data preparation, rows defect units zero, which correspond to cancelled orders were removed as these orders are irrelevant. The defect rate KPI, as mentioned above was derived by dividing the number of defective units by the total quantity delivered. The model was then trained using a combination of numerical and encoded categorical features, including Quantity, Unit Price, Price Difference, Lead Time, Supplier, and Item Category. This structured preparation ensured that the dataset was suitable for regression analysis and that all variables contributed meaningfully to predicting defect outcomes.

The performance of the regression model was then evaluated using two key metrics: Mean Absolute Error (MAE) and the R^2 score. MAE provided a measure of the average magnitude of predicted errors which indicates how closely the model's predicted defect rates aligned with actual defect values. The R^2 score assessed how much variance in defect outcomes was explained by the model. Together, these metrics offered a clear insight into the model's predictive accuracy and its effectiveness in capturing underlying patterns in quality performance of this dataset.

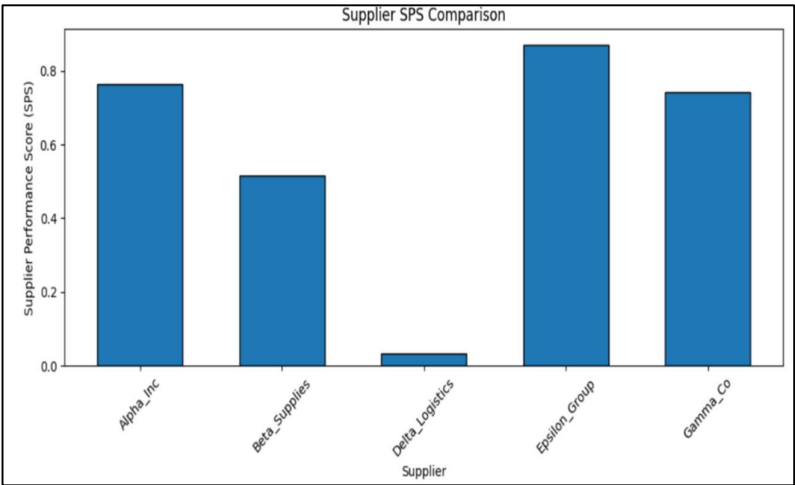
Statistical Results

Descriptive Scoring Analysis

The descriptive analysis using various KPIs that were devised earlier revealed clear performance differences across suppliers. Some suppliers consistently demonstrated stable operational performance, while others showed greater variability and risk.

Suppliers such as Epsilon_Group and Gamma_Co have relatively lower negligence scores, fewer cancellations, and reduced defect rates, supported by their higher Compliance Scores in the KPI summary tables.

In contrast, suppliers such as Delta_Logistics and Beta_Supplies have higher defect rates, greater lead-time variability, and more frequent cancellations which indicate operational instability and higher supply risk. These differences were further reflected in the SPS, where high performers consistently achieved stronger aggregate scores.



When these supplier-level scores were combined with product characteristics to get the procurement recommendation insights, the descriptive analysis also revealed which suppliers were better suited for certain product categories.

For high-criticality categories such as Raw Materials and Electronics, suppliers with strong compliance behaviour and stable lead times, in this case Epsilon_Group and Gamma_Co, emerged as suitable candidates based on their high SPS.

```
prs_pivot = prs_matrix.pivot(index='Supplier', columns='Item_Category', values='PRS')
display(prs_pivot)
```

Item_Category	Electronics	MRO	Office Supplies	Packaging	Raw Materials
Supplier					
Alpha_Inc	3.051906	2.288930	0.762977	1.525953	3.814883
Beta_Supplies	2.066196	1.549647	0.516549	1.033098	2.582745
Delta_Logistics	0.135065	0.101299	0.033766	0.067532	0.168831
Epsilon_Group	3.479656	2.609742	0.869914	1.739828	4.349570
Gamma_Co	2.968547	2.226410	0.742137	1.484273	3.710684

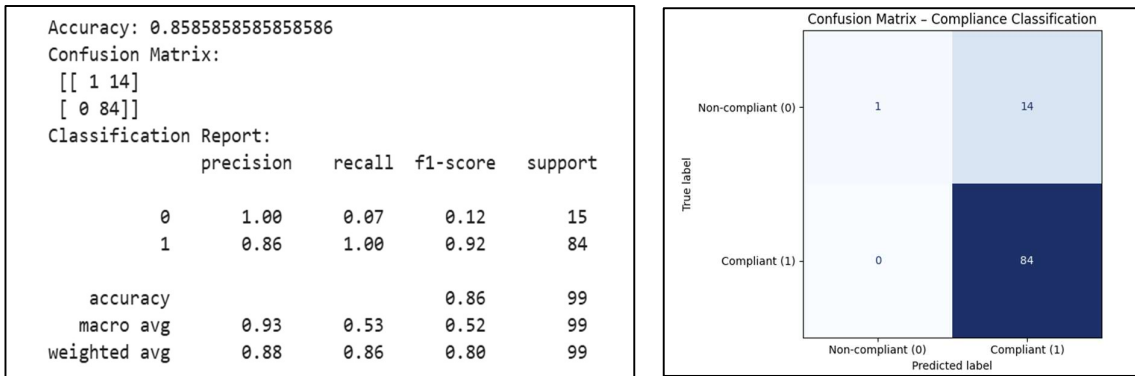
On the other hand, lower-performing suppliers with higher defect rates/ cancellations (Delta_Logistics and Beta_Supplies) were more appropriate for low-risk categories, such as Office Supplies or Packaging, where supply disruptions carry less operational impact.

These descriptive insights provided the foundation for the PRS framework, validating that supplier assignment should be differentiated by both past performance as well as product criticality, which is what US DoD's SPRS is designed to achieve.

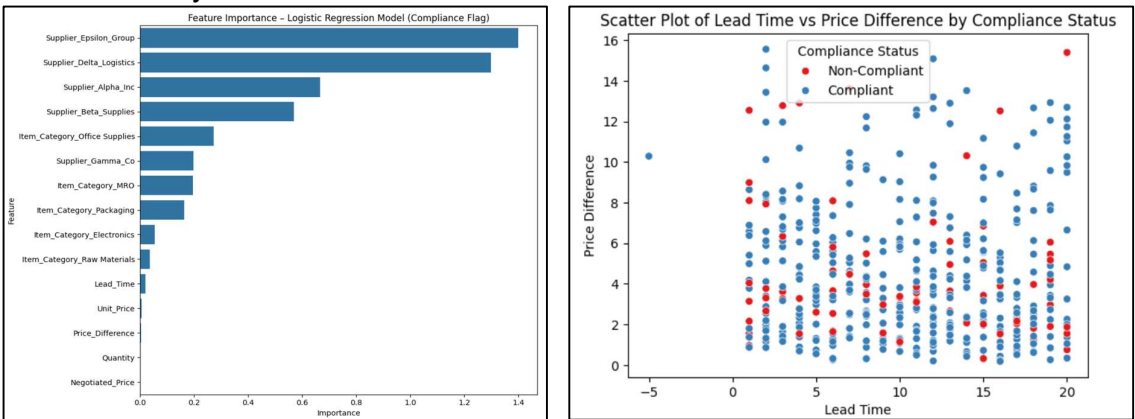
Predictive Model Results

Logistic Regression – Predicting Compliance

The results of the performance metrics (accuracy, precision, recall, and F1-score) from the logistic regression model indicated that the model is reliable in terms of distinguishing between compliant and non-compliant orders.



- Coefficient patterns showed that longer lead times and larger price differences decreased the probability of compliance, while some suppliers and product categories exhibited consistently higher compliance tendencies.
- The recall score remained low for the non-compliant class, which indicated that the model struggled to correctly identify orders at risk of non-compliance.
- The model overall has an accuracy of 85.8% but since there's severe class imbalance (almost all orders are compliant), it performs extremely poorly at detecting class 0 (non-compliant orders).
- Based on the confusion matrix, only 1 out of 15 actual class 0s were predicted correctly. 14 out of 15 class 0s were misclassified.



The scattered distribution of points shows that lead time and price difference alone do not meaningfully differentiate compliant and non-compliant orders, indicating weak

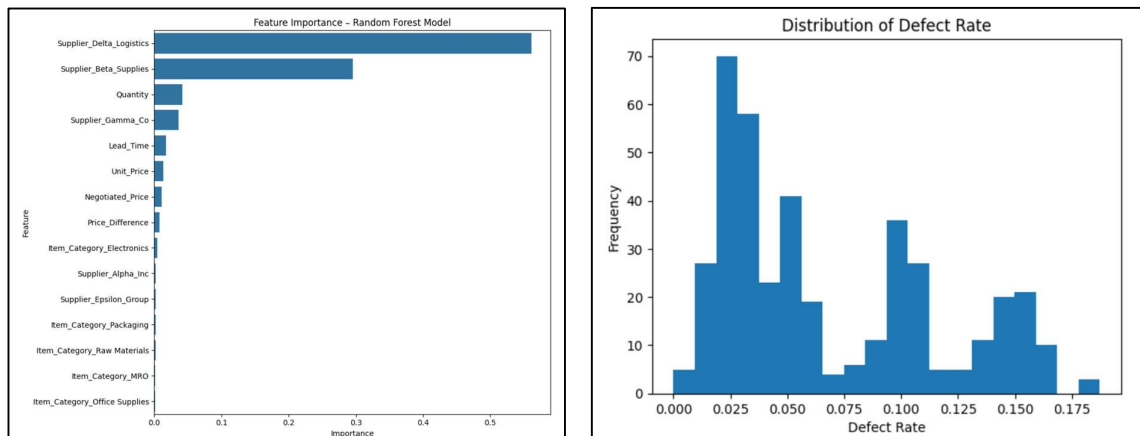
explanatory power for these operational variables. Feature importance results reinforce this, revealing that supplier identity is the strongest determinant of compliance outcomes. Thus, providing enough evidence to reject the null hypothesis and accept the alternate hypothesis that choice of suppliers significantly influences compliance outcomes.

Random Forest Regression – Predicting Defect Rate

The regression model evaluating defect rate performed well according to the Mean Absolute Error (MAE) and R^2 score outputs.

MAE: 0.009594543457727277
R2 Score: 0.917925836909495

- A low MAE indicated that the model's predictions were very close to actual defect values on average.
- The R^2 score demonstrated meaningful explanatory power, confirming that factors such as lead time, quantity, pricing behaviour, and supplier identity contribute to defect outcomes.



Most orders show low defect rates but as per the histogram, a small number of orders have significantly higher defects, indicating maybe the defect rates are supplier-driven. The feature importance results confirm this pattern, as supplier identity is the strongest predictor of defect rate in this case, far outweighing the effect of product categories or operational factors. Thus, we reject the null hypothesis and accept the alternate hypothesis that supplier identity is a prominent driver of defect rates.

This finding further reinforces the need for a Supplier Performance Score (SPS) tailored to the organization's context and grounded in historical supplier data, which enables more informed and sensible procurement decisions.

Conclusions

This study shows that a PRS framework that combines supplier performance, product criticality and predictive modelling offers a practical decision-support tool that can strengthen supplier selection by warning procurement teams of suppliers with consistently lower compliance and defect risks.

The engineered KPIs and the resulting supplier performance index captured these complexities effectively. When this score was combined with product criticality to generate the PRS, the framework enabled strategic supplier-product allocation that takes into consideration quality, reliability, and operational risk.

The predictive models reinforced these findings:

- Compliance risk to a certain extent can be predicted in advance, which allows procurement teams to proactively collaborate with suppliers.
- Defect rate can be forecasted enabling stakeholders to make better decisions on buffer stock.

Together, KPI modelling, SPS, PRS, and predictive analytics form a comprehensive, data-driven procurement decision system. This approach enhances the quality of supplier selection, reduces operational risk, and improves supply chain performance by providing both quantitative and qualitative insights.

Implications in supply chain management

The Supplier Performance Score (SPS) provides a systematic way to distinguish between consistently reliable suppliers and those that pose higher operational risk. By quantifying performance across negligence, defects, cancellations, cost efficiency, and compliance, stakeholders can clearly identify which suppliers are strong candidates for continued partnerships.

When SPS is combined with product criticality ratings, procurement teams can make more informed decisions about whether a supplier is suitable for high-importance items or only to be considered for low-criticality categories. Drawing on the principles outlined by Xia, Ma, and Lim (2008), PRS can be applied in real-life procurement settings if an organization computes their own SPS and PCS values relevant to their supply chains.

Running compliance modelling alongside SPS offers an additional layer of assurance, that allows organizations to predict the likelihood of late or non-compliant deliveries from a supplier. In cases where compliance is low, managers can refer to the defect rate KPI to assess whether quality issues are contributing to reduced delivery performance. They can also run the defect rate modelling to predict the number of units that will be a defected piece and accordingly order buffer stock if cost permits. This insight enables supply chain teams to proactively implement mitigation strategies

and thereby ensure continuity, reducing risk, and strengthening overall supply chain reliability (Pereira, Christopher & Da Silva 2014).

Limitations of the Current Supplier Scoring & Predictive Modelling Framework

This project is subject to several data and methodological limitations. First, key fields such as `Delivery_Date` and `Defective_Units` contain missing values which require assumptions (e.g., treating nulls as supplier negligence or zero defects) which may not reflect true supplier performance.

The dataset includes only five suppliers with relatively few transactions which makes the KPI ratios sensitive to small fluctuations and limiting space for generalization.

Compliance, in this case, is considered heavily reliant on delivery timeliness, oversimplifying a multidimensional concept that typically includes documentation accuracy, service responsiveness, and contractual adherence.

KPI weightages in the Supplier Performance Score were assigned uniformly rather than derived from stakeholder priorities or empirical impact. Product criticality values were manually defined, introducing subjectivity.

The compliance prediction model suffers from significant class imbalance, diminishing its ability to detect non-compliant cases. Finally, the defect-rate regression model, despite high its R^2 , may risk overfitting due to the small dataset and limited feature set, constraining its applicability in the real-world.

In this study, supplier identity was used instead of the engineered KPI variables to keep the models simple, and to avoid KPI-based feature leakage. Supplier identity in this context, therefore, acts as a proxy for deeper behavioral patterns.

In real procurement analytics, predictive models should typically incorporate KPI-level variables or an SPS index directly, offering stronger explanatory power and more realistic supplier-risk insights. This enhancement would have made the present project much more complex beyond its scope.

Future Directions and Improvement Opportunities

Future work can strengthen the PRS framework by incorporating a richer dataset that includes more than 5 suppliers, contractual due dates, and even more granular quality indicators specific to an industry. This would allow for the development of a new KPI focused on Delivery Performance rather than delivery quality being dependent on the compliance flag, as in this case.

Organizations adopting this PRS in real procurement environments should tailor KPI weightages by taking inputs from relevant stakeholders and from analytic hierarchy methods to reflect strategic priorities more accurately. The PRS in this case was

generated for all supply chains that are not military related, essentially civilian supply chains. Therefore, validating PRS across multiple industries or through longitudinal supplier performance data would help assess its generalizability and refine its applicability for long-term procurement decision-making.

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Appendix

Dataset - <https://www.kaggle.com/datasets/shahriarkabir/procurement-kpi-analysis-dataset>

An ipynb file has been uploaded along with the report that shows all the codes and visuals.